

# Lockout-Tagout (LOTO) SOP

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## 1. Purpose

This SOP establishes the procedure for controlling hazardous energy — electrical, mechanical, hydraulic, pneumatic, thermal, and chemical — during the servicing and maintenance of machinery and equipment, in order to prevent unexpected energization, start-up, or release of stored energy that could cause injury or death to personnel.

## 2. Scope

This procedure applies to all employees and contractors who perform installation, maintenance, repair, cleaning, or adjustment of machinery where unexpected energization or release of stored energy could occur. It does not apply to minor tool changes and routine adjustments that are covered under normal production operations and do not require the operator to bypass or remove a guard.

## 3. Definitions

- Lockout: the placement of a lockout device on an energy-isolating device to ensure the equipment cannot be operated until the device is removed.
- Tagout: the placement of a tag on an energy-isolating device to indicate it must not be operated until the tag is removed by the authorized person.
- Authorized Employee: a person who has been trained and certified to perform lockout-tagout procedures.
- Affected Employee: a person whose job requires operation of equipment on which servicing is performed, or who works in the area where such servicing occurs.

## 4. Lockout-Tagout Procedure

- Notify all affected employees that equipment will be shut down and locked out for servicing.
- Shut down the equipment using the normal stopping procedure.
- Isolate all energy sources by operating the disconnect switch, valve, or breaker; do not rely on control switches alone as they do not isolate energy.
- Apply a personal lock and identification tag to each energy-isolating device; each authorized employee performing work must apply their own lock (multi-lock hasp used for group lockout).
- Release, block, or dissipate any stored or residual energy such as spring tension, capacitor charge, elevated components, or pressurized fluid.
- Verify isolation is effective by attempting to start the equipment using normal operating controls, then return controls to the neutral/off position.
- Perform the required servicing or maintenance work.
- On completion, ensure all tools are removed, guards reinstalled, and all employees are clear before removing locks in the reverse order they were applied, and notify affected employees before re-energizing.

## 5. Group Lockout

When multiple crafts or individuals are working on the same equipment, a group lockout procedure using a lockout box or multi-lock hasp is used so that each authorized employee applies an individual lock. Equipment may not be re-energized until every individual lock has been removed by the employee who applied it.

## **6. Removal of Locks by Another Person**

Locks must only be removed by the employee who applied them. In the rare case an employee is unavailable to remove their own lock (e.g., end of shift, emergency), removal may only be performed following the documented exception procedure: verification that the employee is not performing work on the equipment, reasonable attempts to contact the employee, and authorization from the Plant Manager or EHS Head, with the incident logged and the employee informed before their next shift.

## **7. Periodic Inspection**

The EHS department conducts a documented periodic inspection of the lockout-tagout program at least annually for each type of energy control procedure, including observation of an authorized employee performing a lockout to verify compliance with this SOP.

## **8. Training**

Authorized employees receive comprehensive LOTO training covering recognition of hazardous energy sources, procedures for isolation, and the limits of tagout versus lockout devices. Affected employees receive awareness-level training on the purpose of the program and prohibition against attempting to restart locked-out equipment. Retraining is required whenever a new hazard is introduced, a procedure changes, or a periodic inspection reveals deviations.